

Arc Length and Improper Integrals

ANSWER KEY



Arc length

- Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.
 $y' = x^{1/2}$, so $L = \int_0^3 \sqrt{1+x} \, dx = \left[\frac{2}{3}(1+x)^{3/2} \right]_0^3 = \frac{2}{3}(8-1) = \frac{14}{3}$.
- Set up (do not evaluate) the arc length integral for $y = \sin x$ on $[0, \pi]$.
 $L = \int_0^\pi \sqrt{1 + \cos^2 x} \, dx$.

Improper integrals

- Evaluate $\int_1^\infty \frac{1}{x^2} \, dx$, or state that it diverges.
 $\lim_{b \rightarrow \infty} \left[-\frac{1}{x} \right]_1^b = \lim_{b \rightarrow \infty} \left(1 - \frac{1}{b} \right) = 1$: converges to 1.
- Evaluate $\int_1^\infty \frac{1}{x} \, dx$, or state that it diverges.
 $\lim_{b \rightarrow \infty} [\ln x]_1^b = \lim_{b \rightarrow \infty} \ln b = \infty$: diverges.
- Evaluate $\int_0^1 \frac{1}{\sqrt{x}} \, dx$ (improper at the lower limit).
 $\lim_{a \rightarrow 0^+} [2\sqrt{x}]_a^1 = 2 - 0 = 2$: converges to 2.

More arc length practice

- Find the arc length of the line $y = 3x + 1$ from $x = 0$ to $x = 4$, and verify your answer using the distance formula.
 $y' = 3$: $L = \int_0^4 \sqrt{1+9} \, dx = 4\sqrt{10}$. Distance check: endpoints $(0, 1)$ and $(4, 13)$:
 $\sqrt{4^2 + 12^2} = \sqrt{16 + 144} = \sqrt{160} = 4\sqrt{10}$. ✓
- Set up (do not evaluate) the arc length integral for $y = \ln x$ on $[1, e]$.
 $y' = \frac{1}{x}$: $L = \int_1^e \sqrt{1 + \frac{1}{x^2}} \, dx$.

More improper integrals

- Evaluate $\int_1^\infty \frac{1}{x^{1.5}} \, dx$, or state that it diverges.
 $\lim_{b \rightarrow \infty} \left[-2x^{-0.5} \right]_1^b = \lim_{b \rightarrow \infty} \left(2 - \frac{2}{\sqrt{b}} \right) = 2$: converges to 2.
- Evaluate $\int_{-\infty}^0 e^x \, dx$, or state that it diverges.
 $\lim_{a \rightarrow -\infty} [e^x]_a^0 = 1 - \lim_{a \rightarrow -\infty} e^a = 1 - 0 = 1$: converges to 1.

The p-integral test

- 10 For $\int_1^{\infty} \frac{1}{x^p} dx$, state the condition on p for convergence, and explain why using the antiderivative $\frac{x^{1-p}}{1-p}$ (for $p \neq 1$).
Converges when $p > 1$. As $b \rightarrow \infty$, $b^{1-p} \rightarrow 0$ only when $1 - p < 0$, i.e. $p > 1$; otherwise the limit is infinite (or, for $p = 1$, the antiderivative is $\ln x$, which also diverges).
- 11 Use the p -integral result to quickly determine whether $\int_1^{\infty} \frac{1}{x^{0.9}} dx$ converges.
 $p = 0.9 < 1$: diverges.
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Improper integrals with infinite discontinuities inside the interval

- 12 Evaluate $\int_0^2 \frac{1}{(x-1)^2} dx$, or explain why it diverges (note the integrand is undefined at $x = 1$, inside the interval).
Split at $x = 1$: $\int_0^1 \frac{1}{(x-1)^2} dx + \int_1^2 \frac{1}{(x-1)^2} dx$. Near $x = 1$ from either side, $\lim_{b \rightarrow 1} \left[-\frac{1}{x-1}\right]$ blows up to $\pm\infty$: both pieces diverge, so the full integral diverges.
- 13 Evaluate $\int_{-1}^1 \frac{1}{x^{1/3}} dx$ (an odd function with a discontinuity at $x = 0$), splitting at $x = 0$.
By symmetry (odd integrand), $\int_{-1}^0 x^{-1/3} dx = -\int_0^1 x^{-1/3} dx$ (each piece converges since $p = \frac{1}{3} < 1$ for this type of integrand near a finite discontinuity). The two pieces cancel: the integral equals 0.
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Visualizing an improper integral as a limiting area

- 14 The graph shows $f(x) = \frac{1}{x^2}$ on $[1, 10]$. As the right boundary extends to infinity, the shaded area approaches a finite limit. Explain why this is plausible, using the fact that $\int_1^{\infty} \frac{1}{x^2} dx = 1$ (found earlier).
The curve $y = \frac{1}{x^2}$ decreases rapidly toward 0, so the strips added as x grows contribute less and less area. Even though the region extends infinitely far right, the total shaded area converges to a finite value, 1, because the tail contributions shrink fast enough (consistent with the p -test, $p = 2 > 1$).
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Arc length of a more involved curve

- 15 Set up (do not evaluate) the arc length integral for $y = x^2$ on $[0, 2]$, and estimate it using the straight-line distance between the endpoints as a lower bound.
 $y' = 2x$: $L = \int_0^2 \sqrt{1 + 4x^2} dx$. Straight-line distance between $(0, 0)$ and $(2, 4)$:
 $\sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5} \approx 4.47$, which is a lower bound since the actual curve is longer than the chord connecting its endpoints.
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Comparison test for improper integrals

- 16** Use a comparison with $\int_1^{\infty} \frac{1}{x^2} dx$ (known to converge) to argue that $\int_1^{\infty} \frac{1}{x^2+1} dx$ also converges, without evaluating it.
 For $x \geq 1$, $\frac{1}{x^2+1} < \frac{1}{x^2}$. Since $\int_1^{\infty} \frac{1}{x^2} dx$ converges (to 1), and the integrand of interest is smaller and positive everywhere on $[1, \infty)$, the comparison test guarantees $\int_1^{\infty} \frac{1}{x^2+1} dx$ also converges.
- 17** Use a comparison with $\int_1^{\infty} \frac{1}{x} dx$ (known to diverge) to argue that $\int_1^{\infty} \frac{1}{\sqrt{x}} dx$ also diverges.
 For $x \geq 1$, $\frac{1}{\sqrt{x}} \geq \frac{1}{x}$ (since $\sqrt{x} \leq x$). Since $\int_1^{\infty} \frac{1}{x} dx$ diverges and the integrand of interest is even larger, the comparison test guarantees $\int_1^{\infty} \frac{1}{\sqrt{x}} dx$ also diverges.
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A capstone arc-length application

- 18** A cable hangs in the shape $y = \frac{x^2}{8}$ between the towers at $x = -4$ and $x = 4$. Set up the integral for the length of the cable (do not evaluate).
 $y' = \frac{x}{4}$: $L = \int_{-4}^4 \sqrt{1 + \frac{x^2}{16}} dx$.
- 19** Evaluate $\int_2^{\infty} \frac{1}{(x-1)^3} dx$, or state that it diverges.
 $\lim_{b \rightarrow \infty} \left[-\frac{1}{2(x-1)^2} \right]_2^b = \lim_{b \rightarrow \infty} \left(-\frac{1}{2(b-1)^2} + \frac{1}{2} \right) = \frac{1}{2}$: converges to $\frac{1}{2}$.
- 20** A city's water tower is shaped by revolving the curve $y = \sqrt{25 - x^2}$ (the upper half of a circle of radius 5) about the x -axis; set up (do not evaluate) the arc length integral for this curve on $[-4, 4]$, which describes the profile of one side of the tower.
 $y' = \frac{-x}{\sqrt{25 - x^2}}$: $L = \int_{-4}^4 \sqrt{1 + \frac{x^2}{25 - x^2}} dx = \int_{-4}^4 \sqrt{\frac{25}{25 - x^2}} dx = \int_{-4}^4 \frac{5}{\sqrt{25 - x^2}} dx$.